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Population models of sperm-dependent parthenogenesis

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Abstract

Organisms that reproduce by sperm-dependent parthenogenesis are asexual clones that require sperm of a sexual host to initiate egg production, without the genome of the sperm contributing genetic information to the zygote. Although sperm-dependent parthenogenesis has some of the disadvantages of sex (requiring a mate) without the counterbalancing advantages (mixing of parental genotypes), it appears amongst a wide variety of species. We develop initial models for the density-dependent dynamics of animal populations with sperm-dependent parthenogenesis (pseudogamy or gynogenesis), based on the known biology of the common Enchytraeid worm *Lumbricillus lineatus*. Its sperm-dependent parthenogenetic populations are reproductive parasites of the hermaphrodite sexual form. Our logistic models reveal two alternative requirements for coexistence at density-dependent equilibria: (i) If the two forms differ in competitive ability, the form with the lower intrinsic birth rate must be compensated by a more than proportionately lower competitive impact from the other, relative to intraspecific competition, (ii) If the two forms differ in their intrinsic capacity to exploit resources, the sperm-dependent parthenogen must be superior in this respect and must have a lower intrinsic birth rate. In general for crowded environments we expect a sperm-dependent parthenogen to compete strongly for limiting resources with the sexual sibling species. Its competitive impact is likely to be weakened by its genetic uniformity, however, and this may suffice to cancel any advantage of higher intrinsic growth rate obtained from reproductive investment only in egg production. We discuss likely thresholds of coexistence for other sperm-dependent parthenogens. The fish *Poeciliopsis monachalucida* likewise obtains an intrinsic growth advantage from reduced investment in male gametes, and so its persistence is likely to depend on it being a poor competitor. The planarian *Schmidtea polychroa* obtains no such intrinsic benefit because it produces fertile sperm, and its persistence may depend on superior resource exploitation.
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1. Introduction

The evolution and maintenance of sexual reproduction presents a major problem in evolutionary biology because of the costly requirement for males that do not themselves produce offspring (Williams, 1975; Maynard Smith, 1978; Bell, 1987; Hurst and Peck, 1996). Recent modelling of trade-offs between competitive ability and growth capacity have revealed a wide range of conditions for coexistence of competing sexual and asexual sibling species (Doncaster et al., 2000; Kerszberg, 2000; Pound et al., 2002; Doncaster et al., 2003; Pound et al., 2004). These models demonstrate how the two-fold cost in capacity for population growth incurred by the sexual form, due to half its population comprising males, can be compensated by relatively small advantages in competitive ability in crowded environments. The resulting predictions for coexistence of sex with asex provide a time-window for the expression of other evolutionary advantages of sex, such as escape from Muller's ratchet (Kondrashov, 1993; Pound et al., 2004) or from parasites (e.g. Hamilton et al., 1990). The sexual population can even impede the establishment of asexual invaders, if the genetic variation inherent to sexual reproduction confers slight increases in niche dimensions (Pound et al., 2002). A synergy of ecological

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with evolutionary dynamics seems likely to provide a parsimonious explanations for the prevalence in the natural world of sexual reproduction (Pound et al., 2004). Models of the trade-off between growth capacity and competitive impacts can also shed light on the evolution of sexual reproduction from asexual modes. The coexistence thresholds suggest what ecological attributes are necessary for an asexual ancestral population to succumb to descendents with competitive superiority conferred by a sexual mode of reproduction.

The maintenance of asexual reproduction in the presence of sexual modes is an explicit and intriguing problem, in particular for organisms that reproduce by sperm-dependent parthenogenesis, which shows some but not all of the features of sexual reproduction. Sperm-dependent parthenogens are asexual clones that require spermatozoa of a related sexual species to initiate development of eggs, but without the genome of the sperm contributing genetic information to the zygote (Christensen and O'Connor, 1958). The parasitic nature of such individuals binds them into obligatory coexistence with their sexual hosts. A wide variety of organisms exhibit sperm-dependent parthenogenesis (Beukeboom and Vrijenhoek, 1998); at present 24 genera of seven phyla are known to reproduce in this way, and many more are expected to be classified with the advancement of detection methods (Bullini, 1994). This mode of reproduction is also referred to in the literature as animal pseudogamy or plant gynogenesis, and for brevity in this paper we shall often subsequently refer to it as pseudogamy.

The prevalence of such a dependent method of reproduction raises a number of ecological and evolutionary questions (Beukeboom and Vrijenhoek, 1998). It is not clear how sperm-dependent parthenogens arise, nor what stabilizes their coexistence with sexual individuals. From an ecological point of view, their coexistence exhibits simultaneously a parasitic and a competitive aspect. It is parasitic in that sexual individuals waste sperm on the sperm-dependent parthenogens. The latter would be expected to evolve into fully parthenogenetic individuals, however, unless there was some compensation for their obligatory coexistence. In view of the similarity of the sexual and asexual forms in all features but reproductive strategy, they are likely to compete intensively for resources. The principle of competitive exclusion is complicated in this case, however, since the asexual form will impede its own ability to reproduce in the event that it outcompetes the sexual form.

In this paper, we consider ecological models in order to investigate whether predominantly ecological factors are sufficient to help sustain sperm-dependent parthenogenesis. While we seek here to find the most parsimonious explanation for sperm-dependent parthenogens with their sexual hosts, we acknowledge that other factors, such as micro-distribution of sexuals and asexuals or mate choice, could play a significant role.

It has been shown in a number of species, including the salamander *Ambystoma laterale* (Uzzell and Goldblatt, 1967), the bark beetle *Ips accuminatus* (Loyning and Kirkendall, 1996) and the planthopper *Ribautodelphax pungens* (Denbieman and Devrijer, 1987), that sexual forms can distinguish sperm-dependent parthenogens and discriminate against them when mating. This is far from universal however and, in contrast to *R. pungens*, neither of the planthoppers of the genera *Delphacodes* or *Muellerianella* exhibited mate discrimination in laboratory experiments (Booij and Guldemond, 1984; Denbieman and Devrijer, 1987). Increased mating activity of males of the sailfin *molly Poeciliopsis latipinna* increases their attractiveness to females (Schlupp et al., 1994), so that there is a reproductive advantage in mating with an asexual even if the genetic material produced is wasted: as a result no mate choice discrimination is observed. In the case of the freshwater fish *Poeciliopsis lucida* (McKay, 1971) males mated with asexuals if given limited access to females, perhaps for mating experience. Kiester et al. (1981), however, suggest that for many sperm-dependent parthenogens both the assumption that no mate choice is exercised, and that individuals have finite reproductive ability, are approximately true; for example the salamander *Ambystoma maculatum* (Arnold, 1977), the spider beetles *Ptinus* (Sanderson, 1960) and the cankerworm *Alsophila pometaria* (Mitter and Futuyma, 1977).

Differences in micro-distribution of sexual and asexual individuals in the field, in relation to males, have been suggested as the stabilizing mechanism in populations of *Delphacodes* or *Muellerianella*; explicitly, that sexual females may be more efficient at finding males than asexual females (Booij and Guldemond, 1984; Denbieman and Devrijer, 1987). Population subdivision may permit survival by migration and selection (Nagylaki, 1977). Alternatively, sperm-dependent parthenogenesis could be maintained in the absence of stabilizing mechanisms through the creation of new lineages at a rate that keeps pace with the frequent extinction of old lineages.

In general sperm-dependent parthenogens have not been well studied, with the possible exception of the triploid topminnow *Poeciliopsis monacha-lucida* (Vrijenhoek and Pfeiler, 1997) and the freshwater planarian *Schmidtea polychroa* (Weinzierl et al., 1999). Although parasitism is studied extensively in its own right, including for example spatial effects (Ahmed, 1999), multiple host/parasite populations (Hamilton, 1986) and group selection (Levin and Pimentel, 1981), the modelling literature has barely considered sperm-dependent parthenogenetic populations—even in terms of basic ecological dynamics. This type of parasitism is

exceptional in that it may affect reproductive success alone, and not the life cycle of the parasitized individual. The evolution of diploid populations has been considered (e.g. Yi and Lessard, 2000), including quantitative work for diploid and triploid populations (Wu, 1995), but ecological models of coexistence are required to investigate the underlying dynamics.

To our knowledge, the only previous ecological model of sperm-dependent parthenogenesis is a generic analysis by Kiester et al. (1981) for either dioecious or monoecious species with an asexual parthenogenetic sibling species. Sufficient conditions for the extinctions of such systems are given; criteria which are apparently satisfied by the behaviour of a wide range of species considered. Their results indicate that—unless the emergence of sperm-dependent parthenogenesis is always followed by extinction—additional factors beyond those considered in their model play an important part in the population dynamics of such species.

Kiester et al. (1981) do not consider density-dependent dynamics, which is a prevalent force in the natural world that has been shown to be crucially important to understanding ecological costs of sex (e.g. Doncaster et al., 2000), as well as in most other contexts where competition between like populations occurs (only if all populations were predator-limited would one be justified in ignoring density-dependent interactions). In this paper, we demonstrate how sperm-dependent parthenogens can coexist with their sexual hosts in a crowded environment by trading competitive advantage for intrinsic growth capacity. In the following analysis, we consider the conditions required for the persistence, rather than the extinction, of sperm-dependent parthenogens. Classical Lotka–Volterra dynamics are used to develop an ecological model for sperm-dependent parthenogenesis in animals, henceforth referred to as pseudogamy. We show that ecological dynamics alone may be sufficient to sustain coexistence of pseudogamous with sexual subspecies. We concern ourselves with modelling the most general, and hence widely applicable, ecological dynamics. We go on to discuss whether our ecological conditions may suffice for particular species, or whether the maintenance of sperm-dependent parthenogenesis requires additional behavioural responses.

The coupled continuous rate equations for births and deaths of sexual and pseudogamous subspecies allow various degrees of density dependent competition for resources, measured by the Lotka–Volterra coefficient of interspecific relative to intraspecific impacts. A crucial feature of our model is that it accounts for the waste of reproductive resources by the sexual subspecies in initiating pseudogamous reproduction.

We take as a reference species the common Enchytraeid *Lumbricillus lineatus*, which is a small red worm (~ 1 cm) abundant across Europe along the tide-lines of sheltered shores. Enchytraeids in general are of much interest to biologists, with research into both their life cycle and population dynamics (Birkemoe et al., 2000), interactions (Huhta and Viberg, 1999) and self-fertilization (Dozsa-Farkas, 1995), in part due to their potential applications. These include testing for oil contamination (Filimonova and Pokarzhevskii, 2000) and decomposition of waste (Marinissen and Didden, 1997; Edsberg, 2000).

Lumbricillus lineatus is usually a hermaphrodite diploid ($L.\ lineatus_{x2}$), that is, individuals have paired chromosomes, one from each parent. A parthenogenetic (asexual) triploid subspecies ($L.\ lineatus_{x3}$) is also common. It is this subspecies that reproduces pseudogamously, and does not itself produce mature spermatozoa (Christensen and O'Connor, 1958). In addition, there exist tetra- and pentaploid ($L.\ lineatus_{x4,5}$) parthenogenetic forms, which produce genetically unbalanced sperm. Their sperm is sufficient to stimulate reproduction in all asexual *L. lineatus*, but is sterile for sexual reproduction in the diploid form (Christensen, 1980).

Observed populations contain only certain combinations of *L. lineatus* types in significant numbers. The diploid form may of course exist by itself, but is often parasitized by a coexisting triploid population. In the presence of the tetra- or pentaploid form however, the diploid form is very rare or absent, probably because the sperm produced by the asexual forms is fatal for sexual reproduction. Tetra- and pentaploids can exist independently, since they may self-stimulate, and have been found with triploids at both high and low relative densities (Christensen et al., 1976). An important question is why tetra- and pentaploid forms have not invaded all populations of diploids and caused the extinction of sexual forms. In this initial study we consider only the dynamics of the coexisting diploid and triploid forms. Further study of tetra- and pentaploid populations is work in progress.

Our development of ecological models for *L. lineatus* does not explicitly consider genetic variation at this point. Our objective is to explore the ecological conditions necessary for maintenance of pseudogamous triploids in a sexual diploid population. The conditions that sustain stable equilibria are then informative about the adaptations to pseudogamy that may explain its persistence in biological systems through evolutionary time. We expect the models developed here to be relevant also to other species with related processes of reproduction, including vertebrates (e.g. the fish *Poeciliopsis monacha-lucida*) and plants (e.g. the grass *Panicum maximum*). At this stage, however, we do not address any specific changes required to make our models applicable elsewhere.

2. Methods

2.1. Logistic competition

Before proceeding to a model of sexual and pseudogamous sub-species, we briefly remind ourselves of the dynamics of a two-species model that includes competition, but not parasitism (see for example Murray, 1992, Chapter 3). A simple (non-dimensionalized) model of this kind is given by

$$\dot{n}_2 = n_2(1 - (n_2 + \alpha_{23} n_3)), \tag{1a}$$

$$\dot{n}_3 = n_3(\rho - (\alpha_{32} n_2 + n_3)), \tag{1b}$$

where the time derivatives are taken with respect to the natural time scale of subspecies 2; ρ is the ratio r_3/r_2 where r_i is net reproduction rate per capita (birth minus death) of subspecies i, and the competition coefficients α_{ij} measure the inhibitory effect per capita of type j on reproduction by type i relative to the intraspecific impact on reproduction of type i. Then a value of $\alpha_{ij} > 1$ implies an interspecific impact of subspecies j on i that is greater than the intraspecific impact of i on itself. Here we nominate subspecies as 2 and 3 since these will refer to the diploids and triploids, respectively, in later developments of the model.

The important point about this system is that, depending on the values of the competition coefficients and ρ, one may obtain either coexistence of both species or competitive exclusion in which only one subspecies survives. These results are summarized below, and follow directly from study of the phase plane shown in Fig. 1. The four distinct cases are

I. If $\alpha_{32} < \rho < 1/\alpha_{23}$, then all solutions converge on the unique and stable positive (coexistence) steady state.
II. If $\alpha_{32} > \rho > 1/\alpha_{23}$, then the coexistence equilibrium is unstable but both exclusion states, namely (1, 0) and (0, 1), are stable, so that which species survives is determined by initial conditions, with the domains of attraction for each exclusion separated by a separatrix passing through the coexistence equilibrium
III. If $\alpha_{32}\, 1/\alpha_{23} > \rho$, then all solutions tend to (1, 0) giving exclusion of n_3 by n_2;
IV. If $\alpha_{32}\, 1/\alpha_{23} < \rho$, then all solutions tend to (0, 1) giving exclusion of n_2 by n_3.

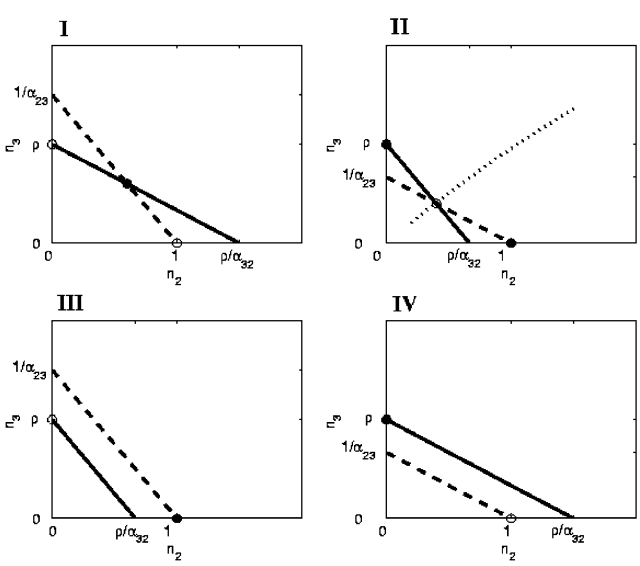

Fig. 1. The null clines of Eq. (1a) (dashed line) and (1b) (solid line) with stable (solid points) and unstable (empty points) equilibria marked in each case: the separatrix is marked by a dotted line. For details of cases (I)–(IV) see text.

Population models of the form given in Eq. (1) are based upon the assumptions of the standard logistic model (Verhulst (1838); see for example. Murray, 1992, Chapter 1). In this case a constant rate of reproduction per capita is offset by a death rate per capita that increases linearly with population density. Net growth per capita consequently declines linearly with density, so that changes over time in the size of a single population can be expressed by

$$\frac{\mathrm{d}N}{\mathrm{d}t} = N\left((b-d) - \frac{N}{C}\right),$$

where b and d are intrinsic rates of birth and death, respectively, and C is a constant. The population grows logistically from an intrinsic net rate $(b-d)$ at low density to zero growth at the carrying capacity, $K = C(b-d)$. The time scale, τ, used in Eq. (1) is set by the intrinsic net rate, with $\tau = (b-d)^{-1}$. The question of evaluating b and d separately therefore does not arise, since only the net reproduction rate is of relevance. This considerably simplifies the analysis, as can be seen by the non-dimensionalized form in Eq. (1) which requires only the ratio of net reproductive rates, ρ.

2.2. Pseudogamous reproduction and the Allee effect

We now consider changes in the basic logistic assumptions necessary to take account of the parasitic effect of the pseudogamous triploids on the population dynamics of the sexual diploids. Pseudogamous reproduction uniquely presents a fundamental difference between intrinsic rates of birth and death that is not present for other modes of reproduction.

A constant per capita rate of reproduction, b, is especially suitable for hermaphrodites such as *L. lineatus* that are unlikely to self-fertilize in nature (Dozsa-Farkas, 1995), but is often also applied in models of two-sex species. Such a model assumes that the success

of reproduction for each individual is unaffected by the population size. However, individuals may experience difficulty in finding a mate in low density populations, in which case we would expect numbers to decrease once the population falls below a certain level—referred to as the Allee effect (Allee, 1931; MaCarthy, 1997). For a hermaphrodite population of size N, this can be represented by a rate of reproduction per capita:

$$\frac{bN}{N+A},$$

where A is a constant which, as part of the population model, defines the 'extinction pit' $(0, N_A)$. Thus $N(t)\to 0$, as $t\to\infty$ whenever $N(t)$ drops below some threshold N_A determined, in part, by A.

The coexistence of a pseudogamous subspecies with the diploid hermaphrodite means that reproduction for individuals from both subspecies requires fertilization by a mate from only one subspecies (the sexual form). Thus the reproductive success of both subspecies depends—in part—on the population of only one. As a first approximation we assume that the individuals of both subspecies choose a mate at random from the entire population. If we let N_2 represent the sexual diploids and N_3 represent the pseudogamous triploids, then either subspecies has a reproduction rate per capita (based on the success rate of finding a suitable mate)

$$\frac{b_i N_2}{N_2+N_3}, \quad i=2,3.$$

Successful reproduction for both diploids and triploids depends on the presence of sperm producing diploids, so that per capita reproduction tends to zero with decreasing diploid numbers. A relatively large triploid population produces an Allee effect, since fewer diploid mates are available for individuals of both subspecies. In the absence of triploids ($N_3=0$), the per capita birth rate of diploids simplifies to the constant b_2, which is what we expect for simple hermaphrodite reproduction. The range of possible outcomes is not qualitatively changed by the inclusion of an additional Allee effect over and above that generated by the presence of pseudogamy (i.e. per capita reproduction equal to $b_i N_2/[N_2+N_3+A]$), and is therefore not considered here. In the full analysis presented in the next section, we will allow for the possibility that diploids and triploids have different intrinsic birth rates, b_2 and b_3, respectively.

2.3. Two-subspecies model

We now pass to a more complete study of the population dynamics of coexistence between pseudogamous and sexual subspecies. We suppose that coexisting diploid and triploid subspecies are bounded by resource limitation acting on reproduction so that each has a density dependent impact on its own population growth. We combine the Allee and competitive effects described in the previous section. The unscaled coupled population dynamics are modelled by the following continuous rate equations:

$$\frac{\mathrm{d}N_2}{\mathrm{d}t} = N_2\left(b_2\frac{N_2}{N_2+N_3} - D - \frac{N_2+\alpha_{23}N_3}{C_2}\right), \tag{2a}$$

$$\frac{\mathrm{d}N_3}{\mathrm{d}t} = N_3\left(b_3\frac{N_2}{N_2+N_3} - D - \frac{\alpha_{32}N_2+N_3}{C_3}\right). \tag{2b}$$

The population consists of N_2 diploids and N_3 triploids. The primitive birth rates of these two subspecies are given by, respectively, b_2 and $b_3=\beta b_2$. The quantity β is thus the ratio of the triploid to the diploid intrinsic birth rates. There is a common per capita death rate D; we observe that for non-trivial population dynamics we require both b_2 and b_3 greater than D, otherwise the diploids or triploids must necessarily go rapidly extinct. The interesting coupled population dynamics therefore only occur in the region $d<1$, β where $d=D/b_2$, since within this region there exists at least the a priori possibility of diploid–triploid coexistence.

The total birth rate in this model is modified by Allee effects as discussed in Section 2.2. The death rates increase with population density in the standard way due to competition (Section 2.1). The diploid population of size C_2 is that size for which the intra-diploid competition-induced death rate just balances the birth rate, and the analogous diploid carrying capacity is $K_2=C_2(b_2-D)$. The quantity C_3 is the size of triploid population at which intra-triploid competition alone would balance triploid birth rate, and the analogous notional triploid carrying capacity is $K_3=C_3(b_3-D)$. It is notional because in the absence of diploids, the triploid birth rate is reduced from its ideal value b_3 to zero. Finally, the α_{ij} are (unsealed) inter-subspecies competition coefficients as defined in Section 2.1.

The non-trivial significance of the death rate complicates the non-dimensionalization of the problem. Here we non-dimensionalize in the following way by setting:

$$n_2=\frac{N_2}{C_2b_2}, \quad n_3=\frac{N_3}{C_3b_2}, \quad \tau=b_2t,$$
$$d=\frac{D}{b_2}, \quad \kappa=\frac{C_3}{C_2}, \quad \beta=\frac{b_3}{b_2},$$

Letting $\dot{n}_2=\mathrm{d}n_2/\mathrm{d}\tau$ and $\dot{n}_3=\mathrm{d}n_3/\mathrm{d}\tau$, Eqs. (2) become

$$\dot{n}_2 = n_2\left(\frac{n_2}{n_2+\kappa n_3} - d - (n_2+\kappa\alpha_{23}n_3)\right), \tag{3a}$$

$$\dot{n}_3 = n_3\left(\beta\frac{n_2}{n_2+\kappa n_3} - d - \left(\frac{1}{\kappa}\alpha_{32}n_2+n_3\right)\right). \tag{3b}$$

We briefly discuss the parameters which enter this scaling. Diploids differ from triploids in this non-dimensionalized model by two parameters, in addition to the competition coefficients. The ratio β is the

intrinsic reproductive rate per capita of triploids relative to that of diploids. The coefficient κ represents the proportional increase in the ability of triploids to exploit their environment, relative to that of the diploids. A value of $\kappa > 1$ reflects the potential for triploids to sustain a higher density of individuals than diploids from a given richness of limiting resource. It is therefore a relative measure of the ability to exploit the environment, all else being equal (MacArthur, 1962; MacArthur and Wilson, 1967). A value of $\kappa > 1$ implies superior equilibrium fitness of the pseudogamous triploids if they have equal competitive ability and intrinsic growth (i.e. if $\alpha_{23} = \alpha_{32} = 1$, $\beta = 1$).

Eqs. (3) have an obvious kinship to conventional two-species competition models with constant intrinsic birth rates per capita (as opposed to rates that vary by $n_2/[n_2 + \kappa n_3]$, e.g. Murray, 1992; Doncaster et al., 2000). With constant rates, competition and carrying capacity measures can be combined by setting

$$a_{23} = \kappa\alpha_{23}, \quad a_{32} = \frac{1}{\kappa}\alpha_{32},$$

since the birth rate is now independent of these effects. As stated above, admissible solutions to the conventional model are coexistence or exclusion by either subspecies of the other, dependent on the competitive impacts a_{23}, a_{32} and, in certain cases, the initial conditions (see Fig. 1). The addition of the Allee effects to the model removes the possibility of a triploid-only equilibrium, since the parasitism inherent in the pseudogamous reproduction mode requires a non-zero diploid population. In the more complete model, we expect rather restrictive constraints on the model parameters for pseudogamous–sexual coexistence.

Below we derive the threshold conditions for coexistence of triploids with diploids under the model given by Eq. (3), by means of differences in either competitive ability or capacity to exploit the environment. For clarity of interpretation, we have separated these mechanisms in the model and consider each in turn. It should be noted that pseudogamy cannot persist in a diploid population with equal competitive ability and equal carrying capacity. This special case, where $\alpha_{23} = \alpha_{32} = 1$, $\kappa = 1$, is analysed in Appendix A and its only outcomes are a pure diploid population or extinction. Likewise, the dynamics of resource renewal cannot sustain coexisting populations in the absence of differences in competitive ability or carrying capacity. Details of the outcomes for a consumer-resource model analogous to the two-subspecies model are also in Appendix A.

3. Results

In the absence of triploids ($n_3 = 0$), Eq. (3) reduces to the classic logistic model discussed in Section 2.1, with diploids dying out if $d > 1$ and otherwise all solutions converging to the steady state

$$n_2^* = (1 - d).$$

In the absence of diploids ($n_2 = 0$), we note that $\dot{n}_3 < 0$ due to the absence of mates, with a resulting extinction of triploids.

3.1. Coexistence mediated through differences in competitive ability

We assume negligible differences in the ability of each species to exploit environmental resources, in order to seek threshold conditions of competitive ability that permit coexistence of triploids with diploids. Eqs. (3) then reduce to

$$\dot{n}_2 = n_2\left(\frac{n_2}{n_2 + n_3} - d - (n_2 + \alpha_{23}n_3)\right), \tag{4a}$$

$$\dot{n}_3 = n_3\left(\frac{\beta n_2}{n_2 + n_3} - d - (\alpha_{32}n_2 + n_3)\right). \tag{4b}$$

Below we consider two specializations of this problem, to investigate the effect of inter and intra species competition. Although system (4) with general d, $\beta > 0$ is much more complex to study analytically, numerics suggest that in practice the qualitative behaviour is described well by the studies below. The possible existence of two positive equilibria, for example, appears to have relatively little impact on the dynamics of the system since one steady state is usually very close to the origin and unstable.

3.1.1. Relatively low intrinsic death rate ($d \ll 1$)

Under the assumption that the intrinsic natural death rate is much lower than the birth rate of diploids ($D \ll b_2$) we consider the system with negligible d. We are therefore interested in the system when death is dominated by intra and inter species competitive effects, rather than intrinsic factors such as age etc.

For $d = 0$, Eqs. (4) have an equilibrium $E_0 = (0, 0)$ representing extinction, and an equilibrium $E_2 = (1,0)$ representing exclusion of triploids by diploids. In addition to these outcomes, certain conditions permit a coexistence steady state $E_* = (n_2^*, n_3^*)$, where

$$n_2^* = \frac{(1 - \alpha_{23}\beta)^2}{(1 - \alpha_{23}\alpha_{32})(1 - \alpha_{32} + \beta(1 - \alpha_{23}))} \quad \text{and}$$

$$n_3^* = \frac{(1 - \alpha_{23}\beta)(\beta - \alpha_{32})}{(1 - \alpha_{23}\alpha_{32})(1 - \alpha_{32} + \beta(1 - \alpha_{23}))}.$$

The origin is critically stable and exhibits a non-empty stability basin. All solutions with sufficiently small initial conditions will converge to E_0, where in this situation it is reasonable to define the size of initial

conditions by the initial per capita diploid reproduction rate $n_2(0)/[n_2(0)+n_3(0)]$. This is determined by the size of n_2 compared to n_3. E_* exists if and only if

$$\left(\alpha_{23}-\frac{1}{\beta}\right)(\alpha_{32}-\beta)>0$$

but is stable only in the case when both brackets are negative. Fig. 2 illustrates the phase plane analysis, with results that may be summarized by the following four cases:

I. If $\alpha_{32}<\beta<1/\alpha_{23}$ then all solutions with sufficiently large initial conditions converge to E_*, coexistence of diploids and triploids;
II. If $\alpha_{32}>\beta>1/\alpha_{23}$ then solutions converge to either E_0 or E_2 depending on initial conditions;
III. If $\alpha_{32}, 1/\alpha_{23}>\beta$ then all solutions sufficiently large initial conditions converge to E_2, the exclusion of triploids by diploids;
IV. If $\alpha_{32}, 1/\alpha_{23}<\beta$ then all solutions converge to E_0, the extinction of both subspecies.

Although extinction may always be attained if $n_2(0)$ is sufficiently small compared to $n_3(0)$, most initial conditions relevant to deterministic modelling will give solution behaviours that are determined by the parameter values as outlined.

We may interpret these solutions as follows. Coexistence of both subspecies is only possible if the type with the lower intrinsic birth rate is compensated by a more than proportionately lower competitive impact from the other, relative to intraspecific competition (Case I). In effect, the intrinsic birth difference β must be neither too large nor too small relative to competitive impacts between forms. Small β leads to exclusion of triploids directly by competitive impact α_{32} from the non-dependent diploids (Case III). Large β results in the dominance of triploids over diploids, leading ultimately to extinction since the pseudogamous triploids rely on the diploids they drive out (Case IV). If the difference in competitive ability is sufficiently mediated by a difference in birth rates (Case II), then exclusion and extinction both have a domain of attraction, with a separatrix passing through the unstable coexistence equilibrium (see Fig. 2). The outcome for Case II is determined by initial conditions, with an initially dominant triploid driving both subspecies to extinction, while an initially dominant diploid population may survive independently. Numerical results suggest that, outside the parameter range defined by IV and II, the extinction equilibrium is unattractive to all but a small subset of initial conditions. In Case II, most solutions go extinct, with only those for which $n_2(0)\gg n_3(0)$ converging to the exclusion state E_2. This situation would, however, be sufficient to prevent any gradual invasion by triploids taking place.

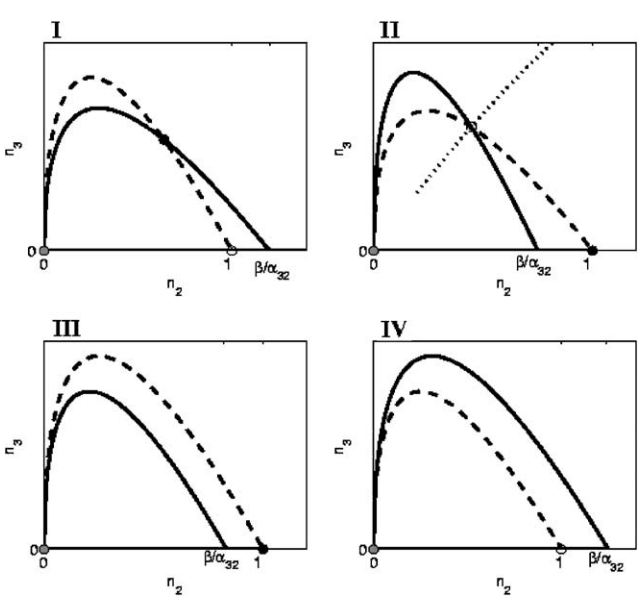

Fig. 2. The null clines of Eq. (4a) (dashed line) and (4b) (solid line) with stable (solid points) and unstable (empty points) equilibria marked in each case: the origin is always critically stable (shaded point). For details of cases (I)–(IV) see text.

It should be noted that the above equilibria are very similar to those of Section 2.1 (Fig. 1), except that here extinction follows inevitably from exclusion of diploids by triploids. In addition, results for the pseudogamous model are not global since sufficiently small initial conditions may also induce extinction. The population dynamics of species when a coexistence solution persists are discussed at the end of in Section 3.3.

3.1.2. *Approximately equal birth rate ($\beta\approx 1$)*

In this section, we present results for the case when $b_2\approx b_3$, so that both species have approximately the same birth rate. The possible cases for $\beta=1$ are given below

I. If $\alpha_{32}<1<1/\alpha_{23}$ and
   i. $(1-\alpha_{23})>(2-\alpha_{23}-\alpha_{32})d$, then all solutions with sufficiently large initial conditions converge to E_*, coexistence of diploids and triploids.
   ii. $(1-\alpha_{23})<(2-\alpha_{23}-\alpha_{32})d$, then all solutions converge to E_0, the extinction of both subspecies.
II. If $\alpha_{32}>1>1/\alpha_{23}$ and
   i. $(1-\alpha_{23})<(2-\alpha_{23}-\alpha_{32})d$ then solutions converge to either E_0 or E_2 depending on initial conditions;
   ii. $(1-\alpha_{23})>(2-\alpha_{23}-\alpha_{32})d$, then all solutions with sufficiently large initial conditions converge to E_2, the exclusion of triploids by diploids;
III. If $\alpha_{32}, 1/\alpha_{23}>1$ then all solutions with sufficiently large initial conditions converge to E_2, the exclusion of triploids by diploids;

IV. If $\alpha_{32}, 1/\alpha_{23} < 1$ then all solutions converge to E_0, the extinction of both subspecies.

Comparison with the results of the previous section shows that the inclusion of non-trivial d only increases the stringency of conditions for coexistence, at least when $\beta = 1$. An increased death rate may, however, switch the population from extinction to exclusion in certain regions of the $[\alpha_{32}, \alpha_{23}]$ parameter space (Case II). The additional condition in Cases I and II determines whether or not a coexistence equilibrium exists: if it does not, then coexistence (requiring a stable equilibrium) is not possible, but neither can the stability basin of the origin extend to a separatrix through any unstable equilibrium. Criteria for sufficiently large initial conditions due to the critical linear stability of the extinction steady state discussed in the previous section are, of course, also required here.

3.2. Coexistence mediated through differences in the ability to exploit resources

The ability to more effectively exploit resources in a given environment is rewarded by a potentially larger population. In this section, we consider the case of negligible difference in competitive ability, but different capacities of each subspecies to exploit the environment. Eqs. (3) become

$$\dot{n}_2 = n_2\left(\frac{n_2}{n_2 + \kappa n_3} - d - (n_2 + \kappa n_3)\right), \tag{5a}$$

$$\dot{n}_3 = n_3\left(\frac{\beta n_2}{n_2 + \kappa n_3} - d - \left(\frac{1}{\kappa} n_2 + n_3\right)\right). \tag{5b}$$

The zero equilibrium $E_0 = (0, 0)$ is critically stable, with a non-empty stability basin, as previously. The exclusion equilibrium $E_2 = (1 - d, 0)$ exists whenever

$$d < 1 \tag{6}$$

and is stable if and only if

$$\kappa(\beta - d) < (1 - d). \tag{7}$$

We henceforth assume that condition (6) is always true, since otherwise extinction is the only outcome. Hence diploids may only exclude triploids if κ and β are both sufficiently small i.e. the triploid does not have big advantages in carrying capacity and birth rate. In addition to these equilibria, a third coexistence steady state $E_* = (n_2^*, n_3^*)$, given by

$$n_2^* = \frac{d^2\kappa(1-\beta)(\kappa-1)}{(\beta\kappa-1)^2},$$

$$n_3^* = \frac{d(1-\beta)(\kappa(\beta-d)-(1-d))}{(\beta\kappa-1)^2}.$$

exists whenever either

$$\beta < 1,\ \kappa > 1 \quad \text{and} \quad \kappa(\beta - d) > (1 - d) \tag{8}$$

or

$$\beta > 1,\ \kappa < 1 \quad \text{and} \quad \kappa(\beta - d) < (1 - d). \tag{9}$$

Note that $\kappa > 1$ in Eq. (8) and $\kappa < 1$ in Eq. (9) become superfluous if we assume $\beta > d$. Although we expect this often to be true, we do not make it a requirement here. Condition (8) implies that E_2 is unstable; if instead Eq. (9) holds, then E_2 is stable. By linearizing the system about the steady state and considering the characteristic polynomial we may show that E_* is stable if and only if

$$(1-\beta)(\kappa-1)(\kappa(\beta-d)-(1-d)) > 0 \tag{10}$$

and

$$(1-\beta)(2d(\kappa-1)^2 + (2-\kappa)(\beta\kappa-1)) > 0. \tag{11}$$

The necessary and sufficient condition for (10) to be satisfied is that Eq. (8) holds. This requires, in addition to the instability of the only other positive equilibrium, that triploids have reproductive disadvantage ($\beta < 1$) but an advantage in their ability to exploit resources ($\kappa > 1$) over the diploids. We therefore consider conditions for Eq. (11) to hold under the assumptions of Eq. (8). We first note that Eq. (8) together with Eq. (6) implies that

$$\beta\kappa > 1 \quad \text{and} \quad \beta > d,$$

the first of which means that Eq. (11) will always hold for

$$\kappa < 2. \tag{12}$$

For $\kappa > 2$ we require that

$$\beta\kappa - 1 < \frac{2d(\kappa-1)^2}{(\kappa-2)}. \tag{13}$$

We may summarize these results as follows:

I. If conditions (6) and (8) hold, and in addition either Eqs. (12) or (13) hold, then all solutions with sufficiently large initial conditions converge to E_*;
II. If conditions (6) and (8) hold but both Eqs. (12) and (13) fail, then we expect solutions with sufficiently large initial conditions to oscillate about E_*;
III. If conditions (6) and (7) hold then all solutions with sufficiently large initial conditions converge to E_2;
IV. Otherwise solutions converge to E_0;

where II follows by Poincaré–Bendixon like argument. The initial condition size is defined as for the competition model. The phase planes are illustrated in Fig. 3. In Case III there are two qualitatively different phase planes: (i) in addition to III, condition (9) holds, so that the coexistence equilibrium E_* exists but is unstable; (ii) condition (9) does not hold and the null clines do not intersect. In both cases the long term behaviour is the same, but the domains of attraction (in particularly,

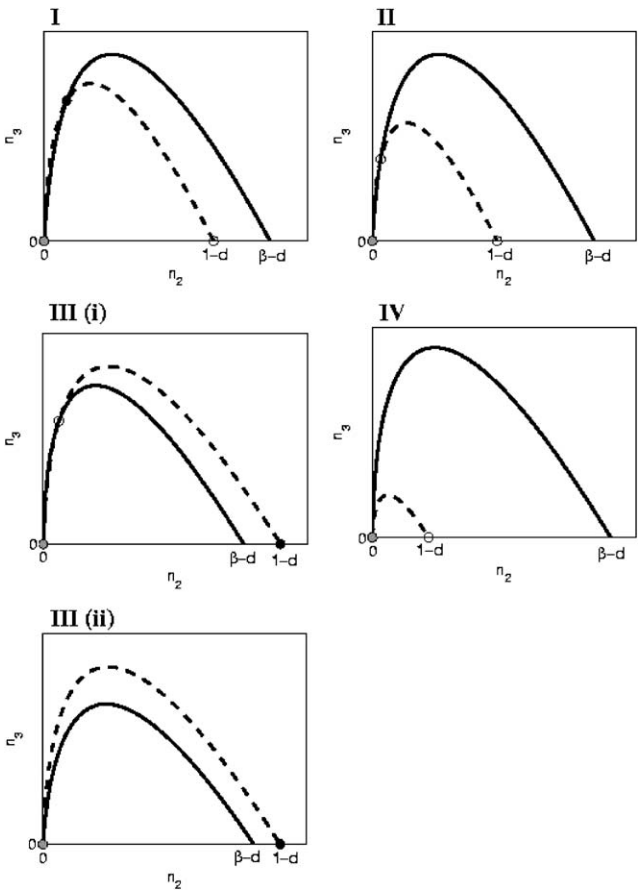

Fig. 3. The null clines of Eq. (5a) (dashed line) and (5b) (solid line) with stable (solid points) and unstable (empty points) equilibria marked in each case: the origin is always critically stable (shaded point). For details of cases (I)–(IV) see text.

what initial conditions are sufficiently large) may be different.

It should be noted that even when E_2 is stable (Case III), the origin may have a large basin of attraction. Under conditions II there exists a positive but unstable equilibrium E_*; so that instead of convergence to a steady state we may instead admit periodic solutions. Contrary to the competition model we therefore generally expect exclusion only if, in addition to conditions (6) and (7) being met (E_2 exists and is stable) we also have $\beta < 1$ or $\kappa > 1$ (i.e. E_* does not exist). The exception to this is initial conditions where $n_2(0) \gg n_3(0)$.

There are therefore very clear biological criteria for the coexistence of pseudogamous triploids. In addition to the obvious requirement that the death rate is not too large ($d < 1$), the triploids must have a lower intrinsic birth rate than the diploids ($\beta < 1$), compensated for by a superior ability to exploit environmental resources ($\kappa > 1$). Conditions (12) and (13) indicate that a strong advantage in resource exploitation will require a significant reproductive disadvantage for steady-state coexistence; otherwise coexistence will be oscillatory. These criteria may, however, be hard to satisfy in practice since a consequence of the existence conditions is that $\beta > d$. Numerical results again suggest that, even in a parameter space satisfying I, initial conditions where triploid populations are too large will drive the system to extinction. Coexistence behaviour is discussed in the next section.

3.3. Coexistence behavour

Although there are clear criteria under which coexistence may occur, the resultant population levels are strongly dependent on all model parameters. In the competition model (Section 3.1) when $d = 0$ and the criteria for coexistence are met, diploids will stay dominant ($n_2^* > n_3^*$) if and only if

$$(1 + \alpha_{32}) > \beta(1 + \alpha_{23})$$

this may or may not be true when the conditions for coexistence are satisfied. Here a difference in intrinsic birth rates may mediate competitive effects to determine the dominant subspecies. In the special case where intrinsic birth rates are equal ($\beta = 1$) conditions reduce, for all $d \geqslant 0$, to the classic requirement that $n_2^* > n_3^*$ if $\alpha_{32} > \alpha_{23}$ i.e. the interspecific impact of subspecies 2 on 3 is greater than that of 3 on 2, while coexistence demands that in addition both interspecific impacts are less than intraspecific ones. Fig. 4 illustrates the two qualitatively different outcomes that may occur when a diploid population at equilibrium E_2 is invaded by a small triploid population (in practice all initial conditions which are not too small converge to the same equilibrium but may have markedly different transient behaviour).

Coexistence mediated by differences in exploitation ability (Section 3.2) require, for $n_2^* > n_3^*$ that

$$\kappa(d\kappa - \beta) + (1 - d) > 0,$$

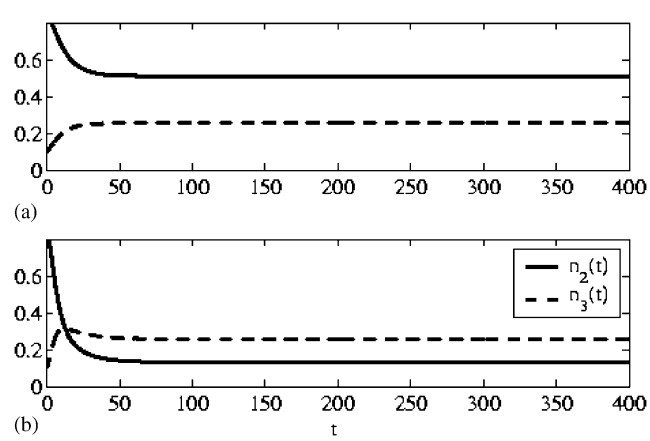

Fig. 4. The solution behaviour for system (5) when $d = 0$, $\beta = 1$ and (a) $\alpha_{23} = 0.8$ and $\alpha_{32} = 0.6$; (b) $\alpha_{23} = 0.6$ and $\alpha_{32} = 0.8$; we have maintenance of a majority diploid population or domination by an invading triploid population, respectively (initial conditions are such that n_2 starts at the exclusion equilibrium E_2 with the invasion of an n_3 population at 10% of this value).

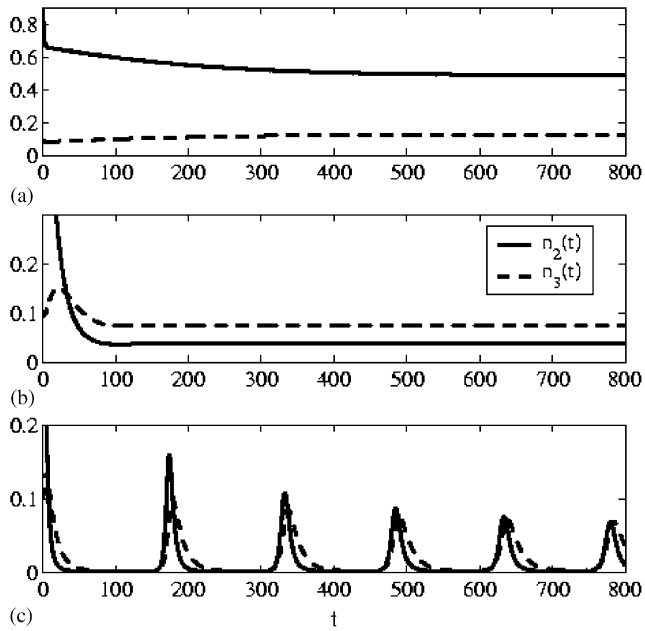

Fig. 5. The solution behaviour for system (6) when $d = 0.1$, $\beta = 0.8$ and (a) $\kappa = 1.2$; (b) $\kappa = 1.5$; (c) $\kappa = 2.5$; results are as for Fig. 4 with the addition of oscillatory solutions, which may again have a mean population of triploids which is larger or smaller (shown here) than that of the diploids.

which may or may not be satisfied regardless of whether E_* is stable. Triploids may therefore dominate the total population ($n_3^* > n_2^*$), or form only a minority ($n_2^* > n_3^*$).

In addition to this, however, the second model also admits oscillatory (coexistence) solutions, converging to a (periodic) limit cycle. Here again either species may be dominant on average. Although in practice each species is usually dominant for part of the cycle, because triploid reproduction is dependent upon diploids population cycles occur synchronically. This is in direct contrast to classic predator prey models, where populations cycle out of phase, and to the competition models discussed above, where no periodic behaviour is possible. For examples see Fig. 5.

4. Discussion

The abundance of pseudogamous subspecies suggests that their existence is not simply an accident or by-product of evolution without adaptive value per se. It remains uncertain, nevertheless, whether such subspecies are evolutionary stable, or merely at some intermediate stage in evolution. Pseudogamy has the clear disadvantage of obligatory coexistence with sexual hosts so that, in the context of *L. lineatus*, triploids will never exclude diploids. Whether the parasitic subspecies derives any benefit from only reproducing (asexually) in the presence of sexual forms is unclear, although their proximity might be a valuable indicator of environmental quality.

In this paper, we have considered simple ecological models of the dynamics of pseudogamy, without explicit reference to genetics. The models combine elements normally associated with competition between species, as well as elements normally associated with parasitism. The presence of pseudogamy has necessitated refining the already fine balance that determines coexistence in the presence of interspecific competition. In particular, even under favourable ecological conditions for the existence of a pseudogamous subspecies, the possibility of extinction is ever-present. Persistence additionally requires favourable initial conditions and lack of environmental fluctuation. Detailed mathematical analysis of the balance between extinction and coexistence is in general complex, and so we postpone it to a more extensive study. Nevertheless, using the simple models presented in this paper, we have been able to show two ecological mechanisms that have the potential to maintain a parasitic pseudogamous subspecies.

The first mechanism for coexistence involves a trade-off between growth capacity and interspecific competitive impacts. As with previous Lotka–Volterra models of sexual types competing with true parthenogens, the relative intrinsic birth rates are crucial to determining competitive outcomes (e.g. Doncaster, 2000; Kerszberg, 2000). For hermaphrodite species such as *L. lineatus*, pseudogamy does not have a built-in cost per triploid to its intrinsic birth rate despite its loss of hermaphrodity. The advantage to hermaphrodite diploids, that only their matings with each other produce zygotes in both individuals of the pair, does not translate into a higher intrinsic birth rate of diploids *per diploid* compared to triploids *per triploid* (assuming no Allee effect over and above that generated by pseudogamy). In contrast, sexual reproduction does have a built-in cost per diploid to its intrinsic birth rate, due to a portion of its reproductive effort being invested in production of male gametes. This is the usual cost of males that applies to sexual forms, whether hermaphrodite or gonochoristic (dioecious), in competition with parthenogens.

We therefore predict $\beta > 1$ for the hermaphrodite *L. lineatus*, as also for gonochoristic species such as the pseudogamous fish *Poeciliopsis monacha-lucida*. Coexistence then requires that the higher intrinsic birth rate of the triploid is balanced by a more than proportionately lower competitive impact on the diploid, relative to intraspecific competition (i.e. $n_2^* > 0$ requires $\beta\alpha_{23} < 1$ from the equilibrium solution to Eq. (4a) with $d = 0$). This may be expected in natural populations, since interspecific impacts are generally weaker than intraspecific, because individuals of the same species are more alike than those of different species. Exactly this condition also applies for competition with true parthenogens: that $n_2^* > 0$ requires $\beta\alpha_{23} < 1$ when d is

set to zero. Thus a long-lived gonochoristic (or dioecious) species with a two-fold cost in reproductive capacity, due to half its output being males that do not themselves reproduce (giving ($\beta = 2$), can only persist if it has a more than two-fold gain in competitive release in the presence of a parthenogen (whether true or pseudogamous).

Recent modelling of true parthenogens has shown, however, that this condition becomes considerably less stringent for organisms with a finite lifespan, given by $d > 0$ (Doncaster et al., 2000; Kerszberg, 2000). Intrinsically short-lived sexual species are predicted to require only a very slight competitive advantage over true parthenogens in order to compensate a two-fold reduction in growth capacity (e.g. Fig. 4a in Doncaster et al., 2000, where a sexual species survives an asexual invasion with $\beta = 2$ and $\alpha_{23} = 0.9$). In such cases the emphasis is on the sexual species persisting after an asexual invasion. For the case of pseudogamy, however, the focus is reversed to find the conditions for persistence of a sperm-dependent parthenogen that can only coexist with the sexual form. Not surprisingly, here we find that $d > 0$ only increases the stringency of conditions for coexistence, at least for $\beta \approx 1$ (Section 3.1.2). Interestingly, if the pseudogamous form evolves into a fully parthenogenetic species, its persistence then depends only on the competitive impact from the sexual form, with any $\alpha_{32} < 1$ favouring persistence if $\beta \geqslant 1$ (Doncaster et al., 2000).

The second mechanism for coexistence requires that triploids have a superior intrinsic capacity to exploit a given resource, compared to the diploids that they pseudogamously parasitize ($\kappa > 1$). The advantage in exploitation must be offset, however, by a relatively low intrinsic birth rate ($\beta < 1$). This seems unlikely for *L. lineatus* in which the triploid invests none of its reproductive effort in seminal vesicles, although some goes into immature spermatozoa and spermathecae (Christensen and O'Connor, 1958) so we expect $2 > \beta > 1$. Values of $\beta < 1$ may apply to other pseudogamous species. For example, the flatworm *Schmidtea polychroa* has a pseudogamous form that produces fertile sperm (Benazzi Lentati, 1970). Its sexual host therefore does not necessarily pay any increased cost of sex, and may have other advantages in intrinsic birth rate resulting from recombination. Coexistence always requires a balance, however, since if either κ becomes too large or β too small then triploids will die out from extinction of diploids or through their exclusion by diploids respectively. This mechanism has dynamics distinct from the competition model since it may produce oscillatory (as well as the more regular steady state) coexistence solutions. Finally, it also indicates the importance of the ability of a species to exploit resources (and in particular the ratio of these for competing species, κ), since here the carrying capacity is only important in terms of its relationship to the measure.

Numerous other organisms are known to have diploid and polyploid subspecies, including plants (Calame and Felber, 2000), molluscs (Qi et al., 2000), marine fish (Felip et al., 2001) and reptiles (Case, 1990). Recent work suggests that these subspecies can have significant fitness differences (Burton and Husband, 2000). Our models predict that triploids can sustain successful parasitism by being relatively poor either in competition or in growth capacity. Given that the genetically uniform triploids can adapt to change only by mutation, and not by recombination of genomes, changeable environments threaten their persistence directly. Stable environments, in contrast, present opportunities for invasion that threaten the persistence of their diploid hosts, and thereby also themselves. We therefore predict that pseudogamous forms are most likely to persist in moderately changeable environments, with pseudogamy arising repeatedly to replace local extinctions. Further models will be required to investigate more complex dynamics, such as reproductive time delays or incomplete mixing of populations (e.g. Weinzierl et al., 1999). These factors could benefit coexistence where a significant triploid superiority is offset by a slow response in the population dynamics, giving oscillatory solutions.

We have shown that simple dynamics based on purely ecological factors are capable of sustaining coexisting populations. However, at this stage of our study, we have deliberately not explored any details of other genetic or geographical factors (such as mate choice or micro-distribution, for example), which may play a significant role in setting low levels of density-dependent competition for some species. One factor in particular, which could in principle modify the detailed conclusions of our study, is the possible existence of species-specific behavioural influence on competition. We remark that, in the absence of this or other stabilizing mechanisms, our study always requires the existence of a competing empty population basin of attraction. This in turn implies infrequent extinction events, as the population jumps probabilistically from stable pseudogamous–sexual coexistence to mutual extinction. Estimates of this frequency require that the underlying stochastic population model be specified precisely. These extinctions require that for long-term population stability there be a balancing mutation process from sexual to pseudogamous forms. We postpone a detailed quantitative discussion of these possibly complex phenomena to a later paper.

In *L. lineatus*, tetra- and pentaploid subspecies coexist with triploids, but their presence generally results in an absence of diploids. Whether this is due to their fitness or because they produce sperm sterile to diploids is unknown. The fact that they are not capable of excluding triploids, which are dependent on—but not necessary to—tetra- and pentaploids, suggests that they

do not have dominant fitness. This is further supported by their distribution, which utilizes only certain subniches of the environment (Christensen et al., 1976). The modelling of such systems is work in progress.

Mathematical explorations of the evolution of sexual reproduction and its maintenance are part of an ongoing debate in population biology. A variety of mechanisms have been proposed, all of which give plausible results under certain circumstances. At present the dynamics and evolution of pseudogamous subspecies has not been fully explored or understood, and the modelling of such subspecies helps to explain their dynamics. Developing mechanisms in models which are capable of sustaining the observed complex evolutionary state and population dynamics of all *L. lineatus* subspecies will give greater insight into the selective forces that contribute to the development and maintenance of sexual behaviour.

Acknowledgements

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Appendix A. Equal competitive and resource exploitation ability

A.1. Two species competition model

In the absence of differences in interspecific competitive ability and carrying capacity, the model is given by the rate equations

$$\dot{n}_2 = n_2\left(\frac{n_2}{n_2+n_3} - d - (n_2+n_3)\right), \tag{A.1}$$

$$\dot{n}_3 = n_3\left(\frac{\beta n_2}{n_2+n_3} - d - (n_2+n_3)\right). \tag{A.2}$$

In this system there exist only two positive equilibria, with an extinction steady state $E_0 = (0,0)$ which is stable whenever the diploid only equilibrium $E_2 = (1-d, 0)$ is not. Clearly E_2 only exists for $d<1$: linear analysis confirms that it is stable if and only if, in addition to this condition, $\beta<1$. Thus diploids exclude triploids if they have a reproductive advantage—if the converse is true, then the superiority of triploids drives diploids (and hence ultimately triploids) to extinction. Extinction can of course be brought about by a sufficiently high death rate $(d>1)$. Once again the coexistence of pseudogamous individuals is not possible, except in the degenerate case $\beta = 1$.

A.2. Consumer-resource model

We consider both subspecies feeding on a single resource. If we let this be represented by some substrate s with logistic growth then $s(t)$ will, in non-dimensionalized form, satisfy

$$\dot{s} = rs(1-s) - s(n_2+n_3), \tag{A.3}$$

where r is the substrate regeneration rate and the second term is the uptake of the substrate by individuals. Letting the reproduction rate of individuals be determined the resource consumed, the population dynamics of diploids and triploids are determined by rates

$$\dot{n}_2 = n_2\left(s\frac{n_2}{n_2+n_3} - d\right), \tag{A.4}$$

$$\dot{n}_3 = n_3\left(s\frac{\beta n_2}{n_2+n_3} - d\right), \tag{A.5}$$

respectively. Our model is therefore defined by system (A.3)–(A.5). Note that we require $\beta \neq 1$ for non-degeneracy.

Below, we show that results for the resource model are analogous to those of the two subspecies system (A.1)–(A.2). It therefore appears that the resource dynamics are not crucial in determining the population behaviour, and suggests that they not significant in producing the population distributions observed in the field.

Steady-state solutions of form (s, n_2, n_3) for system (A.3)–(A.5) are given by $E_0 = (0,0,0)$, $E_1 = (1,0,0)$ and $E_2 = (s^*, n_2^*, 0)$, where

$$s^* = d, \quad n_2^* = (1-d).$$

It can be shown through linear analysis that the origin is always unstable: the local stability of the equilibrium E_1 cannot be determined by this method. Numerical results confirm what can be proven by higher-order methods, that E_1 is stable whenever E_2 is not. The stability of E_2 may be determined by applying the Routh–Hurwitz conditions to the characteristic polynomial of the linearization of (1)–(3) about this equilibrium. These results may be summarized as follows: If d, $\beta<1$, or if $\beta>d>1$, $r>\beta-1$ and $d(\beta-1)^2 > r(d\beta-1)$ then all solutions converge to $E_2 = (s^*, n_2^*, 0)$; otherwise they converge to $E_1 = (1,0,0)$.

Note that extinction of diploids and triploids includes the case $d>1$, β (insufficient reproduction by both) and $\beta>1>d$ (dominance of triploids driving extinction). Persistence of triploids is not possible in this model under any parameter conditions, expect the degenerate case $\beta = 1$, and the system therefore fails to capture the basic dynamics observed in the field.

For the degenerate case $\beta = 1$, our model becomes ill-defined and a whole family of equilibrium solutions exist, in addition to three above, given by (s^+, n_2^+, n_3^+),

where

$$n_2^+ = \frac{rd(1-s^+)}{s^+}, \quad n_3^+ = \frac{n_2^+(s^+-d)}{d}$$

for each $s^+ \in (s^*, 1)$, i.e. for all s between the extinction and exclusion equilibria. Each solution is critically stable and determined by initial conditions (and stochastic effects).

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